Photo by jggrz on pixabay

# Lecture 12

# –

# GMM & EM

## **C-Falter** (*Polygonia c-album*)

- Seinen ungewöhnlichen Namen verdankt der C-Falter seinem einzigartigen Erkennungsmerkmal: Während die Oberseite des Falters orange-braun gemustert ist, zeichnet sich auf der Unterseite der Hinterflügel ein markantes, weißes „C“ ab.
- In Deutschland ist der C-Falter recht häufig zu finden. Dabei fühlt er sich besonders an feuchten Standorten wohl, aber auch Parkanlagen und Gärten besucht er gerne. Er erreicht eine Flügelspannweite von 40-50mm.
- Da er mit Vorliebe den Nektar von Beerensträuchern trinkt, gilt der C-Falter als äußerst guter Bestäuber für diese.

Photo by Daniel Wanke on pixabay

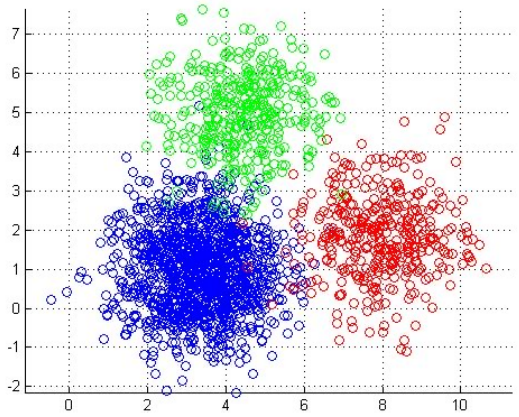
Quellen:
* https://de.wikipedia.org/wiki/Admiral_(Schmetterling)
* https://www.infranken.de/ratgeber/tiere/10-der-schoensten-schmetterlinge-in-deutschland-praechtig-und-bedroht-art-5250577

## Clustering – GMM & EM

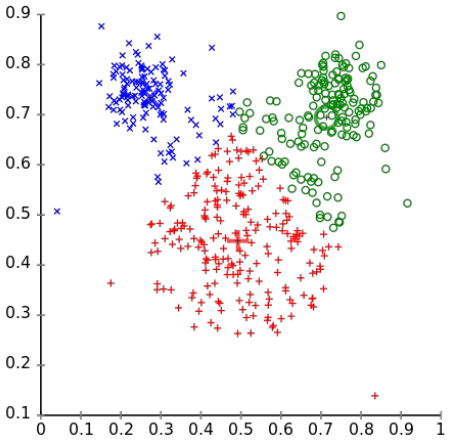
1. From k-means to GMM
2. One-Dimensional GMM and EM
3. Multidimensional GMM and EM
4. Maximum Likelihood Estimation for GMM

# K-means to Gaussian Mixture Models (GMM) (1)

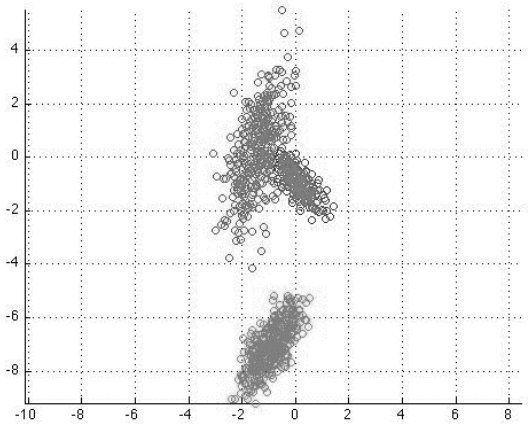
- K-means
  - Each Cluster modelled by cluster center
  - Each data point assigned to nearest Cluster (using Euclidean distance) → „hard clustering“
  - Limitations
    - What about overlapping clusters?
    - What about non-circular clusters?



source: https://de.mathworks.com/



source: https://en.wikipedia.org/wiki/K-means_clustering

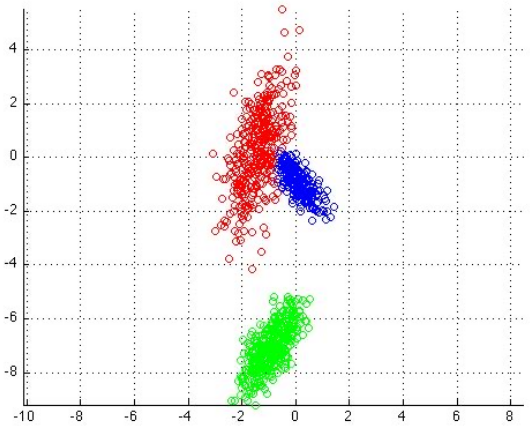


source: https://de.mathworks.com/

# K-means to Gaussian Mixture Models (GMM) (2)

- Gaussian Mixture Model (GMM)
  - models each cluster as a probability distribution, not just a centroid
  - assigns each data point to each Cluster with a probability → „soft clustering“
  - EM-Algorithm → infers mean / (co)variance / weight of each cluster

i.e. how many (%) data points does each cluster contain
(does not have to be evenly distributed)

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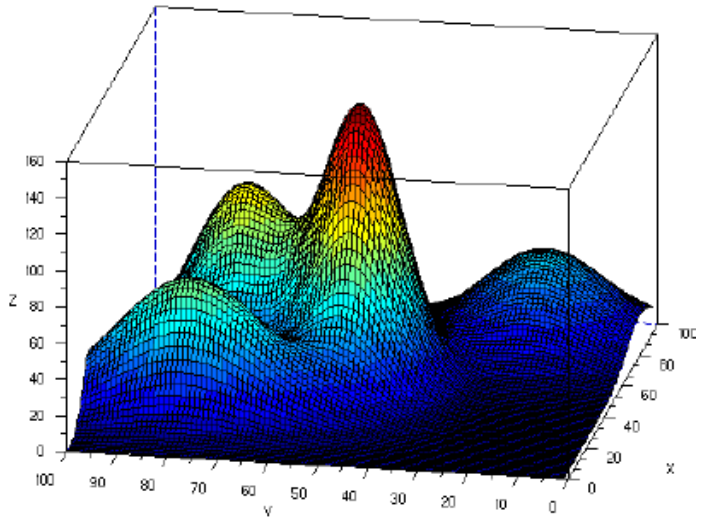



source: https://www.researchgate.net/figure/224105715_fig1_Fig-3-3D-view-of-a-4D-Gaussian-Mixture-Model-used-in-our-experiments

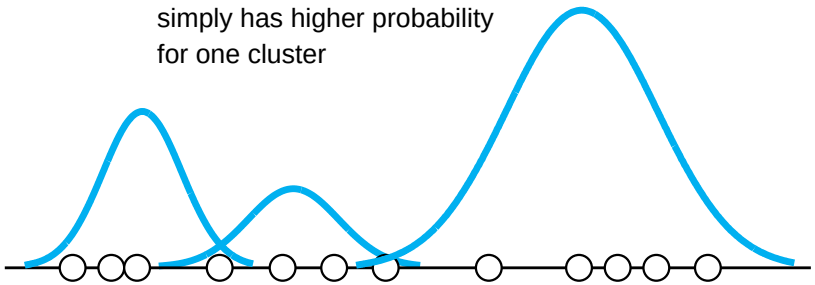
# Gaussian Mixture Model

- Model $k$ clusters as Gaussian distributions with
  - mean $\mu_c$
  - variance $\sigma_c^2$ and
  - weight $\pi_c$ for each cluster $c$ (the percentage of data points generated by $c$ with $\sum_{i=1}^{k} \pi_i = 1$)
- Joint probability distribution: weighted sum $p(x) = \sum_{c=1}^{k} \pi_c \mathcal{N}(x; \mu_c, \sigma_c^2)$

  with $\mathcal{N}(x; \mu, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$ < one-dim gaussian distribution
- „Latent“ form:
  1) select mixture component with probability $\pi$: $p(z = c) = \pi_c$
  2) draw from this components Gaussian: $p(x \mid z = c) = \mathcal{N}(x; \mu_c, \sigma_c^2)$

  ➔ Latent variable $z$: we observe $x$, but $z$ is hidden from us

each data point *will* have a probability for each cluster, so we don't 100% know what cluster it belongs to
simply has higher probability for one cluster



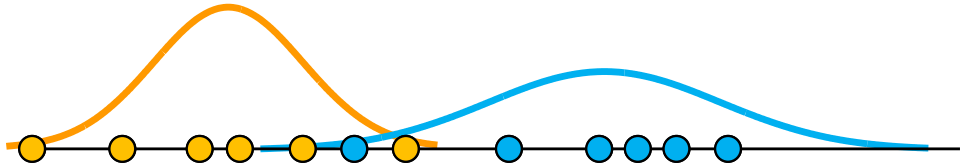
cannot perceive underlying gaussian, would instead be one continous graph since all data points "belong" to all clusters

## Clustering – GMM & EM

1. From k-means to GMM
2. One-Dimensional GMM and EM
3. Multidimensional GMM and EM
4. Maximum Likelihood Estimation for GMM

# One-dimensional Mixture Models

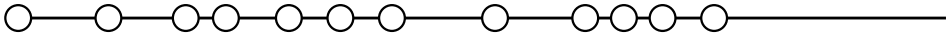
Given $x_1, x_2, \ldots, x_n$ data points



Assumptions 1:

- $k = 2$ Gaussians with unknown parameters $\mu$ and $\sigma^2$
- we know which Gaussians each data points comes from

➔ $\mu_b = \frac{\sum_{i=1}^{n_b} x_i}{n_b}$ and $\sigma_b^2 = \frac{\sum_{i=1}^{n_b} (x_i - \mu_b)^2}{n_b}$ so we could calculate mü and sigma

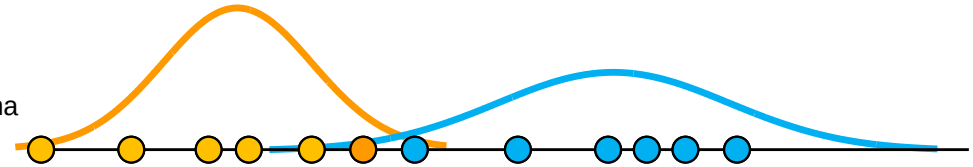


Assumption 2:

- we know the parameters $\mu$ and $\sigma^2$
- unknown which data points come from which Gaussian

➔ $P(b \mid x_i) = \frac{P(x_i|b)P(b)}{P(x_i|b)P(b)+P(x_i|a)P(a)}$ (Bayes Rule) with

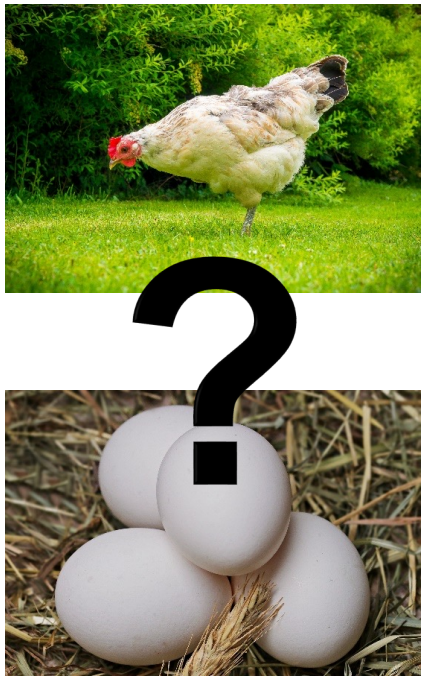
$P(x_i|b) = \mathcal{N}(x_i; \mu_b, \sigma_b^2)$ could calc gaussian from mü and sigma



➔ **Chicken and Egg Problem** if we know one, we could calculate the other but we don't know either

# The EM Algorithm in 1-dim (1)

- Chicken and Egg Problem
  - Need the parameters of the Gaussians to guess the origin of each data point
  - Need the origin of each data point to estimate (max. likelihood) the parameters of the Gaussians



- Solution: The Expectation-Maximization (EM) Algorithm
  - Idea: Start with a random initialization and iterate essentially like k-means

Notation:

- number of clusters $k$ (given)
- mean $\mu_c$, variance $\sigma_c^2$ and weight $\pi_c$ for each cluster $c$
- probability estimation $r_{ic}$ that data point $i$ belongs to cluster $c$

# The EM Algorithm in 1-dim (2)

weight must sum up to 1

- Initialization: randomly choose mean $\mu_c$, variance $\sigma_c^2$ and weight $\pi_c$ for each cluster $c$

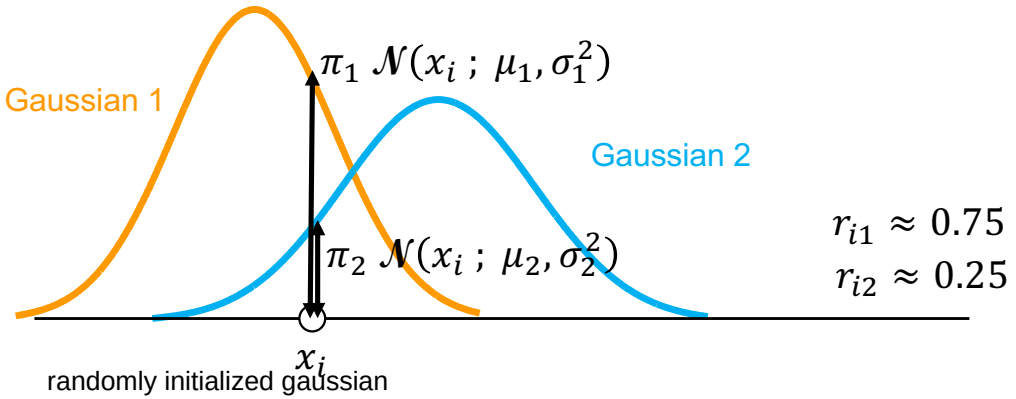
Iterate the E- and M-steps until convergence:

- E-Step: update the probability estimation (responsibility) $r_{ic}$ that $x_i$ belongs to cluster $c$

$$r_{ic} = \frac{\pi_c \, \mathcal{N}(x_i ; \mu_c, \sigma_c^2)}{\sum_{c'=1}^{k} \pi_{c'} \, \mathcal{N}(x_i ; \mu_{c'}, \sigma_{c'}^2)}$$

N is distribution function

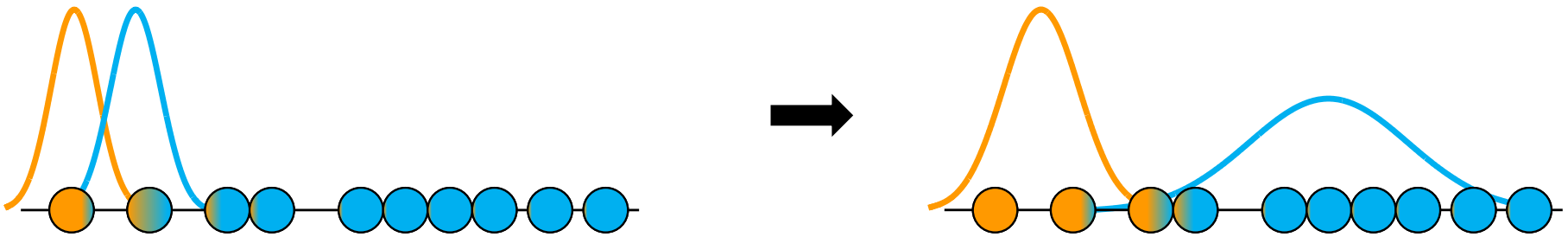
  - probability that data point $x_i$ belongs to cluster c
  - divide by normalization to make the $r$'s sum to 1 for each $x_i$




randomly initialized gaussian

# The EM Algorithm in 1-dim (3)

- M-Step: Maximization: update parameters $\mu_c$, $\sigma_c^2$ and weight $\pi_c$ for each cluster $c$
  - Responsibility of each cluster $c$ : $m_c = \sum_{i=1}^{n} r_{ic}$ sums all r of one cluster
  - Weight of each cluster $c$ : $\pi_c = \frac{m_c}{\sum_{c'=1}^{k} m_{c'}}$ i.e. normalize
  - Weighted mean of each cluster $c$ : $\mu_c = \frac{1}{m_c} \sum_{i=1}^{n} r_{ic}\, x_i$
  - Weighted variance of each cluster $c$ : $\sigma_c^2 = \frac{1}{m_c} \sum_{i=1}^{n} r_{ic}(x_i - \mu_c)^2$ (using the new $\mu_c$ !)





would then iterate further with E-Step and M-Step in sequence
re-calc expectation (cf. above) which essentially moves the gaussian centered around the given points,
> so works like k-means but instead of centroid, gaussian is moved

# EM - Convergence

- Each step increases the (log)likelihood of our model

  this value will always increrase, meaning the gaussians can never move back and return to a previous model

  $$L = \log\big(P(x_1, x_2, \ldots, x_n)\big) = \log\left(\prod_{i=1}^{n} P(x_i)\right) = \sum_{i=1}^{n} \log\left(\sum_{c=1}^{k} \pi_c \mathcal{N}(x_i; \mu_c, \sigma_c^2)\right)$$

  thus, it is guarantueed that the model will converge to one model that is most likely

  Proof: see e.g. https://en.wikipedia.org/wiki/Expectation%E2%80%93maximization_algorithm

- Convergence
  - guaranteed monotonic improvement of likelihood — breaking point is min difference of current L to prev L
    - stop iteration when $L$ (or the parameters) stop changing very much
  - no guarantee of global optimum: may end up in local optimum
    - initialization important ➔ run multiple times, use $L$ to choose best result with different random inits

  advg: can compare L, so models are comparable to find best global optimum

- Hard assignments: assign to most likely cluster

- New points (out-of-training-sample): assign via posterior probabilities

problem: larger k leads to higher likelihood but we want smallest possible k
bc each data point would get a higher cluster probability due to "tighter" clusters
but does not actually represent correct clusters

# Choosing the number of clusters with BIC

- Problem
  - If we increase the number of clusters $k$, the likelihood usually improves
  - But more clusters also mean:
    - more parameters
    - higher model complexity
    - greater risk of overfitting

- Bayesian Information Criterion (BIC): balances model fit & model complexity

$$BIC = -2 \cdot \log L + p \cdot \log n$$

$L$: maximized likelihood of the model
$p$: total number of free model parameters
$n$: number of data points

p stays the same within the same dimensionality
p = 3k -1 for 1-dim

Note: $p$ depends on $k$, the dimension $d$ and the covariance structure

➔ lower BIC = better trade-off between fit and complexity

- How to use BIC
  - fit GMMs with different values of $k$
  - compute the BIC for each model
  - choose the model with the lowest BIC

# Exercises

Exercise 1

1-d EM in Python

## Clustering – GMM & EM

1. From k-means to GMM
2. One-Dimensional GMM and EM
3. Multidimensional GMM and EM
4. Maximum Likelihood Estimation for GMM

# EM for multivariate ($d$-dim) GMMs (1)

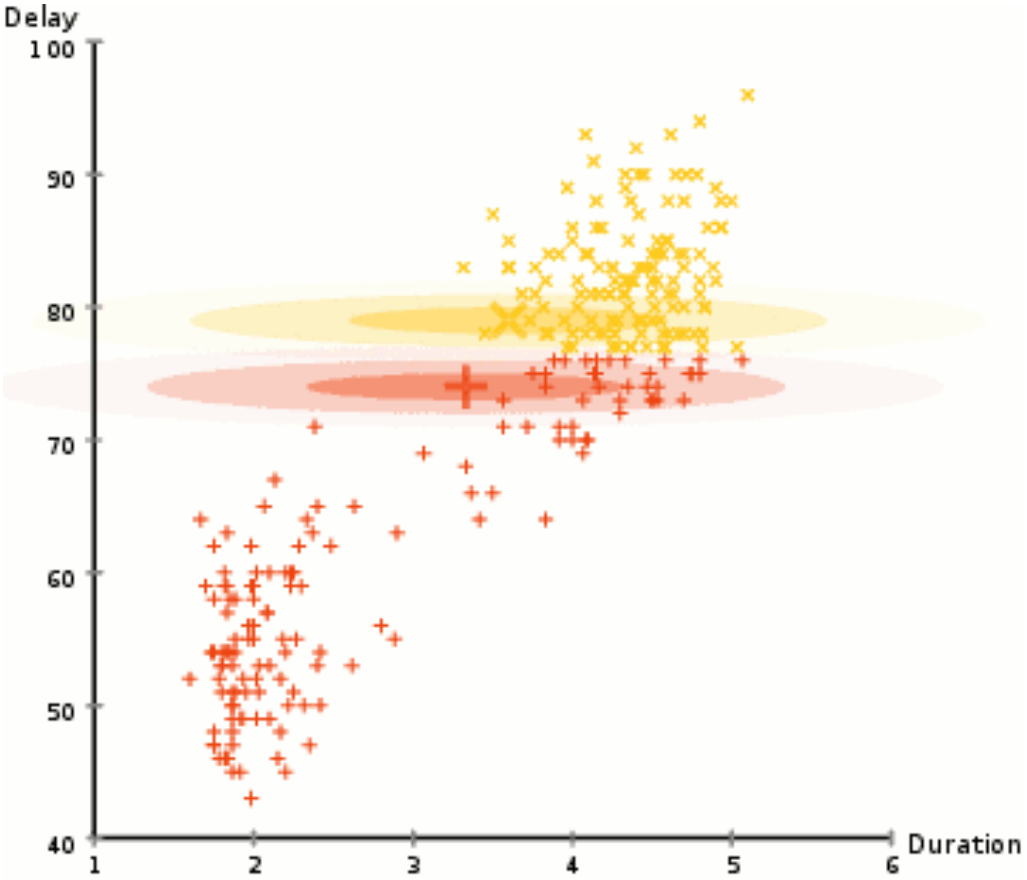
- Data points are $d$-dimensional vectors: $\overrightarrow{x_1}, \overrightarrow{x_2}, \ldots, \overrightarrow{x_n}$
- Mean $\overrightarrow{\mu_c}$ is a $d$-dim vector: $\overrightarrow{\mu_c} = \frac{1}{n} \sum_{i=1}^{n} \overrightarrow{x_i}$
- „Variance“ is a $d$ x $d$-dim Covariance matrix: $\widehat{\Sigma}_c = \frac{1}{n} \sum_{i=1}^{n} (\overrightarrow{x_i} - \overrightarrow{\mu_c})^T (\overrightarrow{x_i} - \overrightarrow{\mu_c})$
- Multivariate Gaussian distribution: $\mathcal{N}(\vec{x}; \vec{\mu}, \widehat{\Sigma}) = \frac{1}{\sqrt{(2\pi)^d \det(\widehat{\Sigma})}} \exp(-\frac{1}{2} (\vec{x} - \vec{\mu})^T \widehat{\Sigma}^{-1} (\vec{x} - \vec{\mu}))$

# EM for multivariate ($d$-dim) GMMs (2)

- E-Step: $r_{ic} = \frac{\pi_c \, \mathcal{N}\left(\overrightarrow{x_i}; \overrightarrow{\mu_c}, \widehat{\Sigma}_c\right)}{\sum_{c'} \pi_{c'} \, \mathcal{N}\left(\overrightarrow{x_i}; \overrightarrow{\mu_{c'}}, \widehat{\Sigma}_{c'}\right)}$ (same but with multivariate Gaussian)

- M-Step:
  - „Responsibility“ of each cluster $c$ : $m_c = \sum_{i=1}^{n} r_{ic}$ (unchanged)
  - Weight of each cluster $c$ : $\pi_c = \frac{m_c}{\sum_{c'=1}^{k} m_{c'}}$ (unchanged)
  - Weighted mean of each cluster $c$ : $\overrightarrow{\mu_c} = \frac{1}{m_c} \sum_{i=1}^{n} r_{ic} \overrightarrow{x_i}$ (same in vector form)
  - Weighted covariance of each cluster $c$ : $\widehat{\Sigma}_c = \frac{1}{m_c} \sum_{i=1}^{n} r_{ic} \, (\overrightarrow{x_i} - \overrightarrow{\mu_c})(\overrightarrow{x_i} - \overrightarrow{\mu_c})^T$

# 2-dim EM Demo



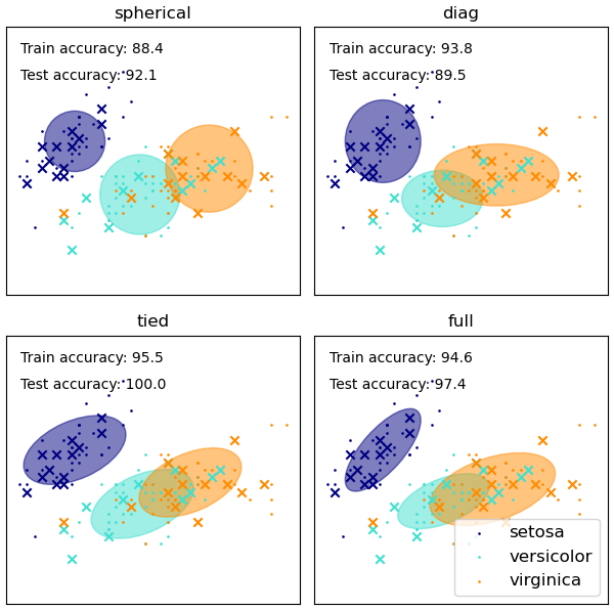

source: https://github.com/lovasoa/expectation-maximization

# Covariance Structures and Parameter Counts in GMM

- Covariance structure determines
  - how flexible each Gaussian cluster is
  - how many parameters the model must estimate

covariance matrix is symmentrical

- A d-dimensional GMM has
  - one mean vector per cluster: d parameters per cluster
  - mixture weights: k-1 free parameters for the whole model
  - one covariance per cluster: depends on the chosen form




- Common covariance structures
  - Spherical: cluster shape: same spread in all directions
    - parameters per cluster: 1 > just k
  - Diagonal: cluster shape: axis-aligned ellipse / ellipsoid
    - parameters per cluster: d > k*d

but must still be an ellipse

  - Full: arbitrary symmetric covariance matrix. Cluster shape: arbitrary ellipse / ellipsoid, including rotation
    - parameters per cluster: d*(d+1)/2 > mult by k
  - Tied: all clusters share the same covariance matrix/shape (but have different means)
    - parameters for the whole model: d*(d+1)/2

all clusters essentially look the same

d refers to dimension

different to prev: is not mult by k bc it's the same covariance for all clusters meaning they use the same params

- More flexible covariance structures can fit the data better, but increase model complexity

source: https://scikit-learn.org/stable/auto_examples/mixture/plot_gmm_covariances.html

# Exercises

Exercise 2

2d-EM Visualization

## Clustering – GMM & EM

1. From k-means to GMM
2. One-Dimensional GMM and EM
3. Multidimensional GMM and EM
4. Maximum Likelihood Estimation for GMM

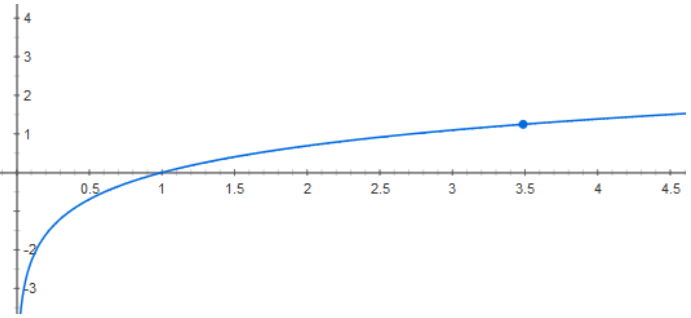
# Estimating the parameters of a 1-d Gaussian Model from observations (1)

- Given $x_1, x_2, \ldots, x_n$ data points (observations) from a 1-d Gaussian Model, estimate the *most likely* parameters $\mu$ and $\sigma^2$ of this Gaussian model
- Def: likelihood of $x_1, x_2, \ldots, x_n$ under the model $\mu$ and $\sigma^2$: $p(\{x_1, x_2, \ldots, x_n\}|\mu, \sigma^2)$
- Maximum Likelihood Estimation (MLE): find $\hat{\mu}, \hat{\sigma} = argmax_{\mu,\sigma} \left(p(\{x_1, x_2, \ldots, x_n\}|\mu, \sigma^2)\right)$
- Assume independence of observations: $p(\{x_1, x_2, \ldots, x_n\}|\mu, \sigma^2) = \prod_{i=1}^{n} p(x_i|\mu, \sigma^2)$
- Take the natural log (monotonically increasing) and use Gaussian

$$\begin{aligned}
\hat{\mu}, \hat{\sigma} &= argmax_{\mu,\sigma} \left(\prod_{i=1}^{n} p(x_i|\mu, \sigma^2)\right) \\
&= argmax_{\mu,\sigma} \left(\ln\left(\prod_{i=1}^{n} p(x_i|\mu, \sigma^2)\right)\right) \\
&= argmax_{\mu,\sigma} \sum_{i=1}^{n} \ln\left(p(x_i|\mu, \sigma^2)\right) \\
&= argmax_{\mu,\sigma} \sum_{i=1}^{n} \ln\left(\frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x_i-\mu)^2}{2\sigma^2}\right)\right)
\end{aligned}$$

ln kürzt exp später raus

annahme: normalverteilung






IID independent identical distribution

# Estimating the parameters of a 1-d Gaussian Model from observations (2)

- Some simple math….

$$\hat{\mu}, \hat{\sigma} = argmax_{\mu,\sigma} \sum_{i=1}^{n} \ln\left(\frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x_i-\mu)^2}{2\sigma^2}\right)\right)$$

$$= argmax_{\mu,\sigma} \sum_{i=1}^{n} \left[\ln(1) - \ln\left(\sqrt{2\pi\sigma^2}\right) - \frac{(x_i-\mu)^2}{2\sigma^2}\right]$$

$$= argmax_{\mu,\sigma} \sum_{i=1}^{n} \left[-\frac{(x_i-\mu)^2}{2\sigma^2} - \ln(\sigma) - \ln(\sqrt{2\pi})\right]$$

$$= argmax_{\mu,\sigma} \sum_{i=1}^{n} \left[-\frac{(x_i-\mu)^2}{2\sigma^2} - \ln(\sigma)\right]$$

$$= argmin_{\mu,\sigma} \sum_{i=1}^{n} \left[\frac{(x_i-\mu)^2}{2\sigma^2} + \ln(\sigma)\right]$$

within argmax/min it is allowed to just leave out constant if they dont change the maximum

# Find the minimum by taking the partial derivatives and setting to 0:

- For $\hat{\mu}$:

$$\frac{\partial \sum_{i=1}^{n}\left[\frac{(x_i-\mu)^2}{2\sigma^2}+\ln(\sigma)\right]}{\partial \mu}=0$$

$$\sum_{i=1}^{n}\left[\frac{-2(x_i-\mu)}{2\sigma^2}\right]=0$$

$$\sum_{i=1}^{n}[x_i-\mu]=0$$

$$\sum_{i=1}^{n}x_i-n\mu=0$$

$$\hat{\mu}=\frac{1}{n}\sum_{i=1}^{n}x_i$$

- For $\hat{\sigma}^2$:

$$\frac{\partial \sum_{i=1}^{n}\left[\frac{(x_i-\hat{\mu})^2}{2\sigma^2}+\ln(\sigma)\right]}{\partial \sigma}=0$$

$$\sum_{i=1}^{n}\left[\frac{-2(x_i-\hat{\mu})^2}{2\sigma^3}+\frac{1}{\sigma}\right]=0$$

$$-\sum_{i=1}^{n}\left[\frac{(x_i-\hat{\mu})^2}{\sigma^3}\right]+\frac{n}{\sigma}=0$$

$$-\sum_{i=1}^{n}[(x_i-\hat{\mu})^2]+n\sigma^2=0$$

$$\hat{\sigma}^2=\frac{1}{n}\sum_{i=1}^{n}(x_i-\hat{\mu})^2$$

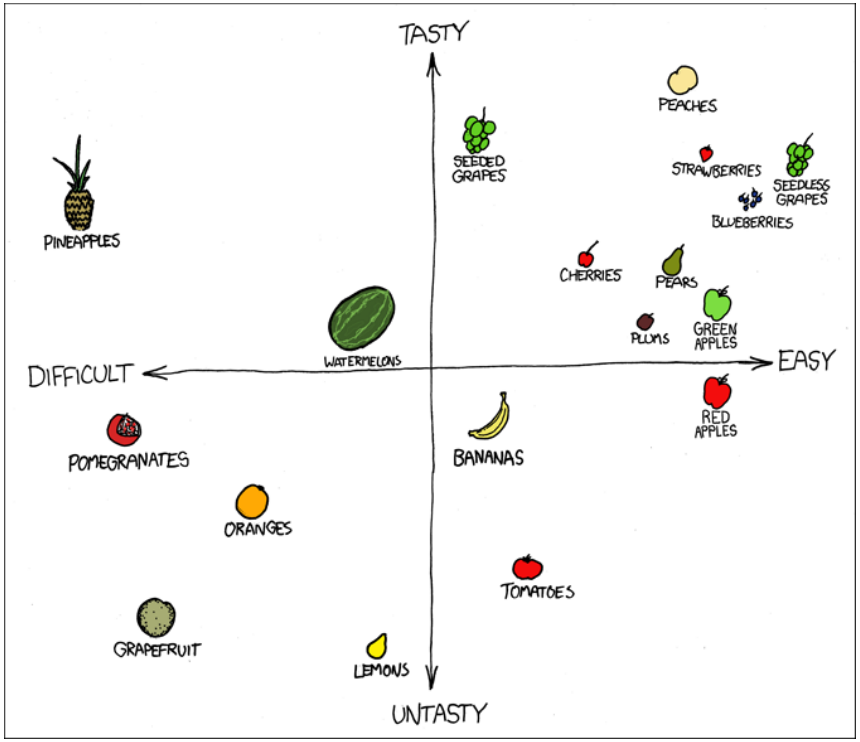
# Estimating the parameters of a $d$-dim Gaussian Model from observations

- Exact same approach as for the 1-dim case, but using
  - $d$-dim vectors $\overrightarrow{x_1}, \overrightarrow{x_2}, \ldots, \overrightarrow{x_n}$ as observation
  - a $d$-dim vector as mean $\hat{\vec{\mu}}$ and
  - a $d$ x $d$-dim covariance matrix $\hat{\Sigma}$
  - and the multivariate Gaussian distribution $\mathcal{N}(\vec{x}; \vec{\mu}, \Sigma) = \frac{1}{\sqrt{(2\pi)^d \det(\Sigma)}} \exp\left[-\frac{1}{2}(\vec{x} - \vec{\mu})^T \Sigma^{-1} (\vec{x} - \vec{\mu})\right]$

leads to

$$\hat{\vec{\mu}} = \frac{1}{n} \sum_{i=1}^{n} \overrightarrow{x_i}$$

$$\hat{\Sigma} = \frac{1}{n} \sum_{i=1}^{n} (\overrightarrow{x_i} - \hat{\vec{\mu}})^T (\overrightarrow{x_i} - \hat{\vec{\mu}})$$




source: https://xkcd.com/388/

## Further Reading:

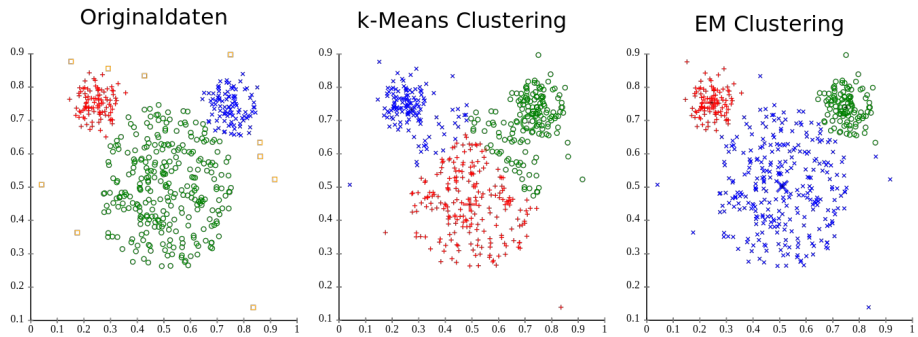
EM and proof it converges
https://en.wikipedia.org/wiki/Expectation%E2%80%93maximization_algorithm
https://www.youtube.com/watch?v=ZZGTuAkF-Hw

MLE:
https://www.youtube.com/watch?v=tV3RIs3NGec
https://www.youtube.com/watch?v=LCWsId9ZCB4

EM:
https://www.youtube.com/watch?v=qMTuMa86NzU
https://www.youtube.com/watch?v=REypj2sy_5U&index=1&list=PLBv09BD7ez_4e9LtmK626Evn1ion6ynrt

# Key Takeaways

- GMM – Gaussian Mixture Models
  - Flexible and frequently used class of probability distributions
  - Explain observations using latent (hidden) groupings of data
- EM Algorithm (Expectation Maximization)
  - Generalization of $k$-means
  - Computes soft cluster membership estimations $r_{ic}$
  - Estimates latent parameters of the Gaussians
  - Various ways to choose number of clusters $k$




source: https://en.wikipedia.org/wiki/K-means_clustering